Shane Patandin

shane.a.patandin@gmail.com | LinkedIn | GitHub | U.S Citizen

Education

Stony Brook University — Stony Brook, NY
Master of Science: Data Science — *Expected December 2026*

Stony Brook University — Stony Brook, NY
Bachelor of Science: Applied Mathematics and Statistics, Economics — GPA: 3.5 — *May 2025*

Relevant Coursework: Statistical Learning, Big Data Systems, Data Management, Data Mining, Machine Learning in Quantitative Finance, Mathematical Statistics, Econometrics

Experience

Artemis Speakeasy — Ronkonkoma, NY
Data Scientist Intern — *May 2026 – Present*

- Engineered a PostgreSQL data warehouse for 17 months of restaurant sales data (∼$2.3M), with automated testing and a 16x faster refresh (96s to 6s).
- Demonstrated that revenue was constrained by demand rather than table turnover, with 29% average occupancy valuing faster turns at ∼$9K/yr instead of the estimated $156K.
- Quantified a ∼14% annual decline in customer volume after adjusting for seasonality, offset in part by an 8% rise in spend per guest driven by menu mix.

Stony Brook University — Stony Brook, NY
Research Assistant — *June 2026 – Present*

- Engineering MetaGlass, an Android application that helps blind users locate and interpret visual charts/graphs in real time using Meta Ray-Ban smart glasses, combining computer vision, sensor fusion, and audio feedback.
- Building a real-time multi-phase pipeline (scan, align, capture, analyze, narrate) that integrates ML Kit object detection and text recognition with a hybrid spatial-tracking system (∼5–15ms latency) to guide users via directional audio cues.
- Integrate proprietary software for multimodal chart analysis, using OpenCV-based pre-cropping and bounding-box refinement to improve description accuracy, with results delivered via text-to-speech narration.

Stony Brook University — Stony Brook, NY
Resident Assistant (RA) — *August 2024 – Present*

- Provided leadership and mentorship to 100+ residents, coordinating programming, communication, and administrative workflows in a high-responsibility environment.
- Managed programming and operational workflows, balancing event coordination, documentation, and policy compliance with attention to detail.

Stony Brook University — Stony Brook, NY
Teaching Assistant (TA) — Applied Linear Algebra — *August 2024 – December 2024*

- Supported professor in delivering course material, helping students understand complex linear algebra concepts such as vector spaces, matrix transformations, eigenvalues, and eigenvectors.
- Assisted in grading 176 assignments and exams, ensuring accurate and timely feedback on student performance.
- Conducted weekly office hours to reinforce course content, address student questions, and provide targeted support for individual learning needs.

Projects

Bank Term Deposit Prediction | GitHub — April 2026 – Present

- Led model fine-tuning and hyperparameter optimization using Optuna across XGBoost, LightGBM, and Random Forest classifiers on highly imbalanced dataset, improving balanced accuracy over untuned baselines.
- Implemented a GPU-trained FT-Transformer with learned categorical embeddings, 8-head self-attention, and validation-based early stopping, achieving the best balanced accuracy among all non-leakage models.
- Designed a leakage-aware preprocessing pipeline using SMOTE and scaling inside CV folds, and demonstrated the inflation effect when using future data.

Skills

Programming Languages: Python, SQL, R, SAS
Libraries: Scikit-learn, PyTorch, Pandas, NumPy, Matplotlib, ggplot2
Databases & Tools: PostgreSQL, SQLAlchemy, Docker, Git, GitHub